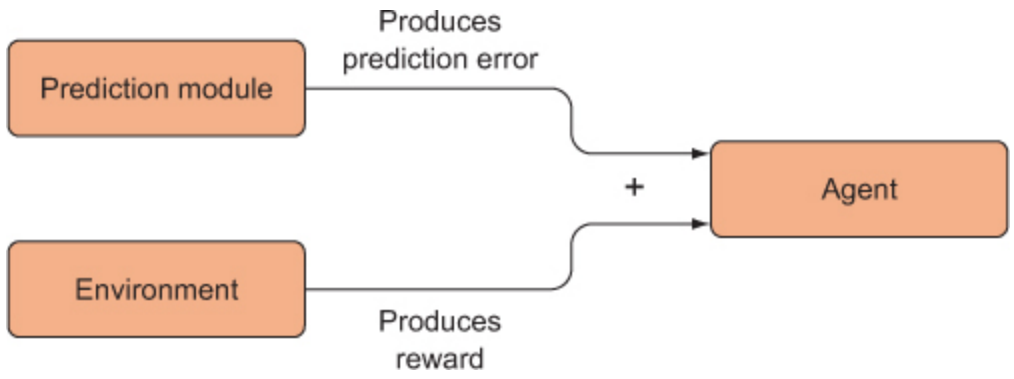
One of the first attempts to imbue reinforcement learning agents with a sense of curiosity involved using a prediction error mechanism. The idea was that in addition to trying to maximize extrinsic (i.e., environment-provided) rewards, the agent would also try to predict the next state of the environment given its action, and it would try to reduce its prediction error. In very familiar areas of an environment, the agent would learn how it works and would have a low prediction error. By using this prediction error as another kind of reward signal, the agent would be incentivized to visit areas of the environment that were novel and unknown. That is, the higher the prediction error is, the more surprising a state is, and therefore the agent should be incentivized to visit these high prediction error states. Figure 8.3 shows the basic framework for this approach.

Figure 8.3. Prediction error is summed with the extrinsic environment reward for use by the agent.



The idea is to sum the prediction error (which we will call the *intrinsic reward*) with the extrinsic reward and use that total as the new reward signal for the environment. Now the agent is incentivized to not only figure out how to maximize the environment reward but also to be curious about the environment. The prediction error is calculated as shown in figure 8.4.

Figure 8.4. The prediction module takes in a state, S_t, (and action a_t, not shown) and produces a prediction for the subsequent state, $\check{S}_{t+1}$ (pronounced "S hat t+1", where the hat symbol suggests an approximation). This prediction, along with the true next state, are passed to a mean-squared error function (or some other error function), which produces the prediction error.